

Facilitating Discussions in Immersive Virtual Meetings with Virtual Agents

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ABSTRACT

This research investigates using virtual agents for facilitating collaboration in immersive virtual meetings. We especially focus on how proactive engagement of virtual agents can improve user experience in discussions held in immersive virtual meetings. We developed an immersive virtual meeting system with a virtual agent that proactively provides feedback on discussions based on the agenda, meeting time left, and talk time of each participant. Findings from a user study comparing passive and proactive virtual agent behaviours suggest proactive behaviours can help facilitating an immersive virtual meeting yet could be distracting, needing further investigation to provide more nuanced and relevant intervention.

Index Terms: Virtual meeting, virtual agent, group work.

1 INTRODUCTION

The use of virtual meeting technology is becoming popular, from desktop video conferencing (e.g., Zoom) to fully immersive Virtual Reality (VR) systems (e.g., VRChat, Horizon Workplace) [1]. With rise of Large Language Models (LLMs), intelligent Virtual Agents (VAs) are used in VR for various applications, such as enabling human users to have natural conversations with virtual characters [3]. This research investigates how VAs could be useful in Immersive Virtual Meetings (IVMs) by facilitating group discussions.

Previous research investigated how robots can facilitate face-to-face meetings [2] or LLMs can guide social interaction in the real world [4]. However, less research has been conducted on the use of VAs to facilitate discussions in IVMs. Compared to previous work, we especially focus on how proactive behaviour of VAs in an IVM can improve user experience through facilitating discussions.

2 SYSTEM DESIGN AND IMPLEMENTATION

We designed and implemented a proof-of-concept IVM system that allows multiple participants to meet in VR where a VA facilitates discussion (see Fig. 1). The system allows both verbal and non-verbal communication between participants through voice chat and body gestures made through user's avatars. It also supports a VA that facilitates discussions between participants. The VA plays the role of facilitator in an IVM, announcing and tracking the agenda for the discussions, keeping the meeting time, and encouraging participants who have contributed less to the discussions.

The prototype system is implemented based on the Unity 3D engine (<https://unity.com> v6000.0.63f1) with its VR multi-user template. It supports cloud based multiplayer services, stylised avatars with head and hand gestures, and voice chat features to setup basic collaborative VR environment. We integrated Meta's speech-to-text solution (<https://wit.ai>) to transcribe the voice chat between users, which is then provided to a cloud based LLM service (<https://gemini.google.com>) that drives the VA's behaviour for facilitating discussions between human participants.

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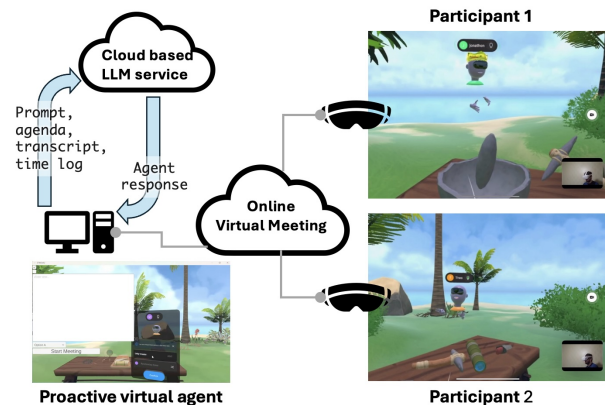


Figure 1: An IVM system that supports an LLM based intelligent VA facilitating discussions between human participants.

While the system is capable of using a human-like avatar to represent the VA, our current research focus is on verbal intervention, hence we decided to represent the VA as a simple virtual sphere to avoid bias and misunderstanding from showing human-like appearance. The virtual sphere provides basic visual cues for user engagement by changing its colour (initially grey) when it is preparing for response (dark blue) and while talking (light blue).

The VA shows basic behaviours of starting the meeting by announcing the agenda, answering user's questions when called with a key phrase (e.g., "Hey, Mediator!"), and making concluding remarks at the end. We also prompted the LLM to generate proactive interventions to encourage less engaged participants to talk, remind the agenda when discussions go astray, and warn when time is short. This prompt is sent to the LLM regularly along with the agenda, a transcript of the conversation, and the talk time of each participant from the speech-to-text module. The LLM returns a suggested text message for the intervention with the likelihood of needing it. The LLM response is stored temporarily, and the system waits until it detects silence longer than a threshold (e.g., 3 seconds), then plays the latest message from LLM using text-to-speech.

3 USER STUDY

To evaluate the system, a user study was conducted after approval by the research ethics committee of the host institute.

3.1 Study Design

The user study compared two variants of the VA behaviour: Passive (P) and Active (A). In Passive condition (the baseline), the VA showed limited participation in the meeting, merely announcing the agenda at the beginning, closing the discussions at the end, and passively answering users' questions. In Active condition, in addition to the baseline behaviour, the VA also proactively facilitated discussions by encouraging participants who spoke less, reminding participants of the agenda when discussions went astray, and warning

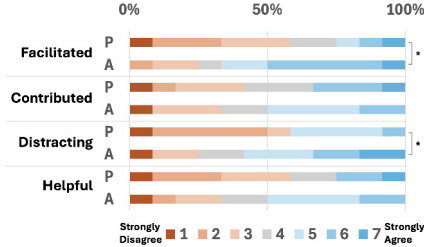


Figure 2: Comparison of ratings on VA in each condition (P: Passive / A: Active / *: significant difference).

them when time was running out.

Participants were recruited in pairs and asked to use the system under each condition. The study followed a within-subjects design with counterbalanced conditions. The study started with participants signing a consent form and answering a demographic questionnaire. A brief training followed to let the participants become familiar with the VR headset (Quest 3) and the controller for navigation and object manipulation in VR. The participants were then tasked to discuss on a meeting agenda under each condition. The meeting agenda was to discuss and rank a list of items based on how useful they would be if the participants were lost on a desert island. Participants were asked to rank items on a table (e.g., an axe, a compass, a telescope, etc.) for a survival task. There were two sets of seven items randomly assigned to two tasks (catching or cooking a fish). The tasks were assigned to the conditions in a counterbalanced manner. Participants were given 5 minutes to discuss the topic in each condition. After completing discussions under a condition, participants answered a set of rating questions (1: strongly disagree - 7: strongly agree) on how much the VA ‘facilitated the discussions’, ‘meaningfully contributed to the meeting’, ‘was distracting’, and ‘was helpful’. After completing both conditions, the participants provided preference between the two conditions and answered a set of rating questions on how useful (1: very useless - 7: very useful) each type of VA behaviour is. Participants were also asked open-ended questions about the reason for their preference and what could be improved in the prototype system. Overall, the study took less than an hour for each pair.

3.2 Results and Discussions

We recruited 12 participants (5 female, 7 male) aged between 23 and 39 years. All participants had prior experience using desktop online meeting systems and VR headsets. Most participants never (58%) or rarely (33%) used VR meeting systems before. All participants used LLM services, at least a few times a month.

The results of the rating questionnaire are summarised in Fig. 2. Wilcoxon Signed Rank tests ($\alpha=.05$) indicated that the participants perceived that the VA facilitated the discussions significantly more ($Z=-2.02$, $p=.043$) in the Active condition ($Mean=4.9$, $Median=5.5$, $Inter-QuartileRange=[3.5-6]$) than in the Passive condition ($M=3.5$, $Md=3$, $IQR=[2-4.5]$). However, the VA in the Active condition ($M=4.7$, $Md=5$, $IQR=[3.5-6]$) was also significantly ($Z=-2.12$, $p=.034$) more distracting than the Passive condition ($M=3.3$, $Md=2.5$, $IQR=[2-5]$). No significant differences were found on how much the VA contributed or how helpful it was.

Although the Active VA was perceived as facilitating discussions more, the majority of the participants (58%) chose the Passive condition as their preferred option, while three participants chose Active and two reported no preference between the two. Participants explained their preference by noting that the Active VA was “interrupting discussion” and, from time to time, “annoying” when they preferred not to speak. The two participants who preferred the Ac-

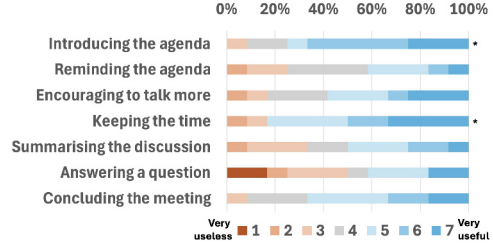


Figure 3: Ratings on the usefulness of various virtual agent behaviours. (*: significant difference from mid-point).

tive VA explained that interacting with the Passive agent required more effort, needing to call out the VA each time for engagement.

Among the various behaviours, ‘introducing the agenda’ and ‘time keeping’ were rated as the most useful, followed by ‘making conclusions’ and ‘encouraging to talk more’ (see Fig. 3). One-sample Wilcoxon Signed Rank tests showed that the participants rated significantly higher than the mid-point (4) for ‘introducing the agenda’ ($Z=2.62$, $p=.009$) and ‘time keeping’ ($Z=2.26$, $p=.024$).

The findings indicate that VAs with proactive behaviours are better at facilitating the meeting, yet they are also distracting. Participants’ feedback indicates that the length of discussions was brief, hence the proactive behaviours were perceived as unnecessary and interrupting from time to time. This suggests that it needs to be tested in longer discussions in real-life scenarios. User preference indicates that proactive behaviours need to be more subtle and timely. Feedback from participants suggests that VAs could utilise non-verbal communication cues (e.g., vocal tone or facial expressions) to decide when to intervene with more relevant suggestions.

4 CONCLUSION

Through the development of a proof of concept prototype and conducting a user study, this research investigated how a VA can contribute to facilitating IVMs. Through a user study, we found that proactive behaviours of a VA can be useful for facilitating IVMs, yet they are also distracting, needing more nuanced and subtle intervention. Future work needs to explore techniques for enabling VAs to recognise and express non-verbal communication cues, and provide more timely and favourable suggestions in an IVM.

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